Mini Assessment

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1. If an API returns 429, what should your program do?

Answer: A `429 Too Many Requests` HTTP response code indicates that the client has exceeded the API rate limits within a given time frame. When this occurs, the program should:

- Inspect the response headers for a `Retry-After` directive, which specifies the exact duration to pause before attempting another request.
- If no header is provided, implement an exponential backoff strategy with randomized jitter to systematically increase delay intervals between successive retry attempts.
- Enforce a maximum retry limit to avoid infinite loops, gracefully aborting and alerting or logging the failure if the threshold is reached.

2. If response.json() fails, what should your program do?

Answer: A failure in `response.json()` typically raises a `JSONDecodeError` (or ValueError), usually caused by an empty body, HTML error pages, or malformed data. The program should:

- Wrap the decoding call within a `try-except` block to prevent abrupt application crashes.
- Verify the HTTP status code (e.g., `response.status_code`) and check the `Content-Type` header to confirm whether JSON was actually returned.
- Inspect raw content using `response.text` to capture helpful diagnostics, log the raw payload along with error details, and raise a clean, handled application exception or fallback response.

3. If a JSON key is missing, why might .get() be safer than []?

Answer: Accessing a dictionary via bracket notation like `data['key']` raises a fatal `KeyError` if the key is absent in the payload. In contrast:

- The `.get()` method (e.g., `data.get('key')`) safely returns `None` instead of raising an unhandled exception.
- It allows defining a custom default fallback value via `data.get('key', 'default_value')`, ensuring downstream operations remain stable.
- It significantly improves code resilience and cleanliness when working with dynamic, nested, or optional fields in third-party API payloads.